

Practical No-03: Autoencoder and Variational Autoencoder for Image Reconstruction

Submitted by-
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1.Github repository: [GitHub](#)

2.COLAB LINK:-

https://colab.research.google.com/drive/1y3FBKmQpPHYBQV15_zYah0H2MY14kvc7

Title: Image Reconstruction and Generation using MNIST

Aim: To implement and compare an Autoencoder and Variational Autoencoder (VAE) for image reconstruction and generation using the MNIST dataset.

Objectives:

- Understand Autoencoder architecture.
- Understand VAE architecture and latent space.
- Implement image reconstruction using both models.
- Compare their reconstruction performance.
- Visualize the learned latent representations.

Dataset: MNIST handwritten digit dataset

Results:

- Autoencoder Reconstruction MSE: 0.002008
- VAE Reconstruction MSE: 0.038625
- Generated images and VAE 2D latent-space visualization were obtained.

Conclusion: The Autoencoder provides effective image reconstruction and feature learning, while the VAE learns a probabilistic latent space that supports both reconstruction and generation of new images.